

MongoDB Aggregation – \$lookup Operator (Hinglish Guide)

1. \$lookup ka kaam

- Ye ek collection ke documents ko doosri collection ke sath join karta hai.
- SQL ke JOIN jaisa kaam karta hai.

2. Basic Syntax

```
{ $lookup: { from: "dusriCollection", localField: "currentCollectionField", foreignField: "dusriCollectionField", as: "nayaField" } }
```

3. Example 1: Simple Join

Orders Collection:

```
{ _id: 1, item: "Pen", customerId: 101 }  
{ _id: 2, item: "Book", customerId: 102 }
```

Customers Collection:

```
{ cust_id: 101, name: "Rahul", city: "Delhi" }  
{ cust_id: 102, name: "Priya", city: "Mumbai" }
```

Query:

```
db.orders.aggregate([  
  {  
    $lookup: {  
      from: "customers",  
      localField: "customerId",  
      foreignField: "cust_id",  
      as: "customerDetails"  
    }  
  }  
])
```

Result:

```
{  
  _id: 1,  
  item: "Pen",  
  customerId: 101,  
  customerDetails: [ { cust_id: 101, name: "Rahul", city: "Delhi" } ]  
}
```

4. Example 2: \$unwind ke sath

- \$lookup ka result hamesha array hota hai.
- Agar ek hi object chahiye toh \$unwind use karte hain.

```
db.orders.aggregate([  
  {  
    $lookup: {  
      from: "customers",  
      localField: "customerId",  
      foreignField: "cust_id",  
      as: "customerDetails"  
    }  
  },  
  { $unwind: "$customerDetails" }
```

```
)
```

5. Example 3: \$lookup with Pipeline (Advanced)

```
db.orders.aggregate([
{
  $lookup: {
    from: "customers",
    let: { custId: "$customerId" },
    pipeline: [
      { $match: { $expr: { $eq: ["$cust_id", "$$custId"] } } },
      { $project: { name: 1, city: 1, _id: 0 } }
    ],
    as: "customerDetails"
  }
}
])
```

6. Real-life Use Cases

- E-commerce: Orders ke sath customer aur product details lana.
- School DB: Student ke sath subjects aur teachers ka join.
- Blogging: Posts ke sath author aur comments lana.

Summary

- \$lookup = SQL JOIN jaisa.
- Result array hota hai, single object ke liye \$unwind karo.
- Filtering/projection ke liye pipeline option use karo.